The mass of any substance while heating remains the same however its volume increases. Thus, its density decreases. But water exhibits an anomalous expansion on cooling and contraction on heating within a specific range of temperature. When water at 0°C is heated, it is observed that it contracts between 0°C to 4°C, i.e. volume decreases and thus density increases. The volume of water is minimum at 4°C. Hence the density of water is maximum at 4°C.

Optics

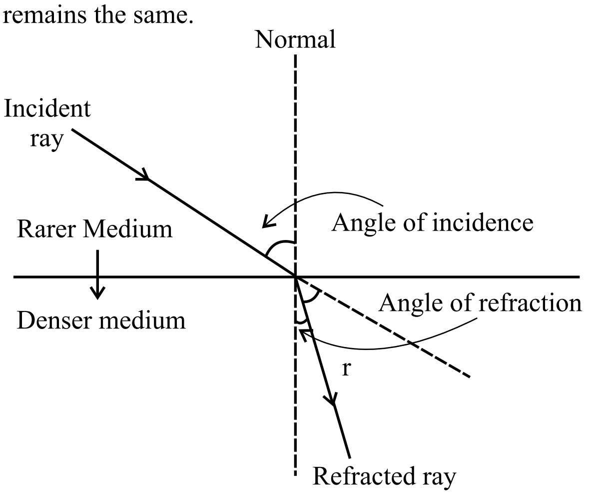
1. When light enters glass from air, its wavelength

(a) first increases and then decreases
(b) increases
(c) decreases
(d) does not change

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

When the light enters from a rarer medium to denser medium, the speed of the light decreases but the frequency remains the same.



As we know,

Speed of Light (v) = Frequency of light (n) × Wavelength (λ)

if $\left\{\begin{array}{l} \downarrow = \text{constant} \times \downarrow \\ \uparrow = \text{constant} \times \uparrow \end{array}\right\}$

Hence, when light enters glass from air, its wavelength decreases.

2. What type of lens is used in magnifying glass?

(a) Concave lens (b) Plano-concave lens
(c) Convex lens (d) Convex mirror
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

Convex lens is used in magnifying glass.

3. A person standing in front of a mirror finds his image bigger than himself. This implies that the mirror is-

(a) Concave
(b) Plane
(c) Convex
(d) Cylindrical with bulging outwards
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

With a concave mirror, a person can get his image bigger than himself.

4. In a barber's shop, two plane mirrors are placed

(a) parallel to each other
(b) perpendicular to each other
(c) at an angle of 45 degree
(d) at an angle of 60 degree
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

The opposite walls of a barber shop are covered by plane mirrors placed parallel to each other, so that infinite images of both front and back are formed in both the mirrors.

5. The visible range of solar radiation is-

(a) 100-400 nm (b) 400-700 nm
(c) 740-10000 nm (d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

The electromagnetic spectrum encompasses all type of radiations. The part of the spectrum that reaches earth from the sun is between 100 nm to 10^6 nm. This band is broken into three ranges - Infrared (above 700 nm), Visible (400 to 700 nm), and Ultraviolet (below 400 nm).

6. When a soap bubble is charged, its radius

(a) becomes zero (b) increases
(c) decreases (d) does not change

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

Whether the bubble is given negative charge or positive charge, the radius will increase in both cases because when positive charge is given to it, again the charges will repel each other and this will expand the bubble and radius will increase. It happened in both cases because of the ionic similarities between the charges.

7. **The wavelength of visible spectrum is in the range :**
   (a) 1300 Å - 3900 Å (b) 3900 Å - 7600 Å
   (c) 7800 Å - 8200 Å (d) 8500 Å - 9800 Å
   (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

Visible light is that part of electromagnetic radiation which can be seen by human eyes. Visible light is usually defined as having wavelengths in the range of 400-700 nm (4000-7000Å), between the infrared and ultraviolet. A typical human eye will respond to wavelengths from about 380 to 750 nm. Thus, among the given options, option (b) will be the right answer.

8. **Which of the following has higher energy level and shorter wavelength?**
   (a) Infrared radiation (b) Ultraviolet radiation
   (c) Visible radiation (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Ultraviolet (UV) radiation covers the wavelength range of 100–400 nm, which is a higher frequency and shorter wavelength.

9. **In big showrooms and departmental stores, the vigilance mirror used is a**
   (a) plane mirror
   (b) convex mirror
   (c) concave mirror
   (d) combination of concave mirror and convex mirror
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

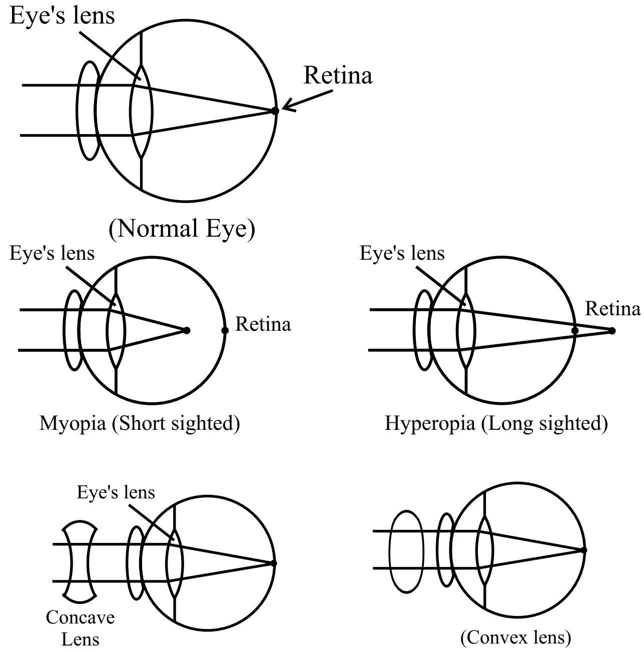
A convex mirror is a curved mirror where the reflective surface bulges out towards the light source and the inner surface is polished.Convex mirrors show a large field of view.The image formed by a convex mirror is erect, virtual and diminished.So, the image of a wide range of area is virtually formed by the convex mirror.Thus, convex mirrors provide the ability to see around corners which is enormously valuable when your aim is to protect your shop from thieves. Therefore, convex mirrors are used as vigilance mirrors in shopping malls.

10. **Myopia can be corrected by using a _____ lens.**
   (a) cylindrical (b) concave
   (c) convex (d) bifocal

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

In a myopia eye, the image is formed in front of the Retina, so a diversing lens is used to shift the image to retina, that's why myopia can be corrected by using a concave lens of a suitable power.



11. **The type of mirror used in vehicles to see the traffic on the rear side is**
   (a) plane (b) concave
   (c) convex (d) None of the above

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Convex mirrors are used in vehicles to see the traffic on the rear side because they always give an erect image and wide field of view as they are curved outwards.
This mirror is also called diverging mirror because it generally diverges the beam of light after reflection.

12. **The focal length of normal eye lens is about**
   (a) 1 mm
   (b) 25 cm
   (c) 2 cm
   (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

The focal length of a normal eye lens is about 2 cm because distance between eye lens and retina is approximately 2 cm. So, when parallel rays are coming they form image at focus and to make image at retina focal length should be about 2 cm.

13. The nature of radiation of light is-

(a) Like wave
(b) Like particle
(c) Like both of wave and particle
(d) Like neither of wave nor of particle

42nd B.P.S.C. (Pre) 1997-98

Ans. (c)

The light has a dual nature, it behaves like both wave and particles. In the later part of the 19th century and in the beginning of 20th century, it was realized that black body radiation and the photoelectric effect can be understood only on the basis of particle model of light. Some experiments require light to be a wave, while others require light to be a particle. This led to the acceptance of dual nature of light. Quantum mechanics explains the duality of light by describing it as a wave-packet. A wave-packet refers to waves that may interact either as spatially localized, acting as particle, or interacting like waves.

14. Which type of waves are light waves?

(a) Transverse waves (b) Longitudinal waves
(c) Electromagnetic waves (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Light is an energy form with a dual nature. Light has both particle and wave qualities implying that it has both particle and wave aspects.
By the law of electromagnetic light is an electromagnetic wave and it is also a transverse wave where the movement of the particular to the propagation of the wave.

15. Which of the following photoelectric devices is most suitable for digital applications?

(a) Photodiode (b) Photovoltaic cell
(c) Photoemitter (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Photovoltaic cells, also known as solar cells, play a crucial role in digital technology, with a significant impact on Satellite Communication Systems. These cells serve as primary power source for satellites, making them vital for the successful operation of space based Communication Networks.
Photo electric effect–The phenomenon where the emission of electrons occurs when a material is exposed to electromagnetic radiation, such as light. Devices that utilise the photoelectric effect are designed to correct light energy into electricity or electric signals. One such device is the photovoltaic cell.

16. Solar cell is a device that converts light energy into electrical energy. We obtain the highest output of energy during noon when the sun is just above our head, because

(a) the density of light radiation is maximum when Sun rays are falling vertically
(b) the heat produced is highest in the noon
(c) the sun is nearest to the earth at noon
(d) None of the above

BPSC Agriculture Department-2024

Ans. (a)

A solar cell is a device that converts light energy into electrical energy. We obtain the highest output of energy during noon when the Sun is just above our head because the density of light radiation is maximum when sun rays are falling vertically.

17. A boy stands at 30 cm distance in front of a mirror. He sees his direct image whose height is $\frac{1}{5}$ of his actual height. Which mirror does he use?

(a) Flat (b) Convex
(c) Concave (d) Plano-concave
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

A boy sees his image in the mirror which is one-fifth of his real image and the distance between the mirror and the boy is 30cm. So the mirror must be a 'Convex Mirror' according to the given condition.
- This is the reason why the side mirrors of vehicles are Convex Mirrors, which can cover more area as it reduces the size of the object.

18. The speed of light in vacuum is nearly –

(a) 3×10^{10} meter/sec (b) 3×10^{8} meter/sec
(c) 3×10^{8} km/sec (d) 3×10^{8} light years

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The speed of light in vacuum is 3.00×10^{8} m/s, while the speed of sound in vacuum is zero and in air is 343 m/sec. The speed of light in glass is 2.0×10^{8} m/s. The velocity of light always greater than the velocity of sound. The velocity of celestial objects and the rockets is quite lower than the velocity of light.

19. The time taken to reach the Sunlight up to the surface of earth is approximately –

(a) 4.2 sec (b) 4.8 sec

(c) 8.5 minutes (d) 3.6 hrs.

42nd B.P.S.C. (Pre) 1997

Ans. (c)

The sunlight takes about 499 second or about 8.2 minutes to reach to the earth. Thus, option (c) is the right answer.

20. The sunlight from the sun to the earth reaches in :

(a) 5 minutes (b) 6 minutes

(c) 8 minutes (d) 10 minutes

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

See the explanation of the above question.

21. Which of the following does not change when light travels from one medium to another?

(a) Velocity (b) Wavelength

(c) Frequency (d) Refractive index

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

When light travels from one medium to another, the frequency of the light does not change, while its velocity, wavelength and refractive index of the medium are changed.

22. Monochromatic light enters from one medium to the other. Which one of the following properties does not change?

(a) Frequency

(b) Amplitude

(c) Velocity

(d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (a)

Monochromatic light enters from one medium to the other medium then the frequency of light remains constant but its amplitude, velocity and wavelength changes.

23. Choose the correct statement.

(a) Wavelength of red light is less than violet light

(b) Wavelength of red light is more than violet light

(c) Wavelength of violet light is more than green light

(d) Wavelength of violet light is more than yellow light

(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

In a visible light spectrum, VIBGYOR (Violet, Indigo, Blue Green, Yellow, Orange and Red) is the order of the increasing wavelength of the various coloured lights where violet coloured light has the minimum and red coloured light has the maximum wavelength. Similarly, the same (VIBGYOR) is the order of decreasing frequency where violet light has the maximum and red light has the minimum frequency.

24. Sky is blue because –

(a) Blue colour in the sunlight is more than other colours

(b) Short waves are scattered more than long waves by atmosphere

(c) Blue colour is more absorbing to eyes

(d) Atmosphere absorbs long wavelength more than short wavelength

39th B.P.S.C. (Pre) 1994

Ans. (b)

The blue appearance of the sky is due to scattering of sunlight from the atmosphere. When we look at the sky, it is the scattered light which enters the eyes. Among the shorter wavelengths, the blue colour is present in a large proportion in sunlight. Light of shorter wavelength is scattered by air molecules because of their smaller size follow Rayleigh's scattering. Blue light is strongly scattered by the air molecules and reaches to the observer. This explains the blue colour of the sky.

25. A beam of blue light was used to obtain diffraction pattern using a diffraction grating. If yellow light is used by replacing the blue light

(a) diffraction pattern becomes narrower and crowded together

(b) diffraction pattern does not change

(c) diffraction pattern becomes broader and farther apart

(d) None of the above

BPSC Agriculture Department-2024

Ans. (c)

If blue light is replaced by yellow light in a diffraction pattern, the amount of diffraction will increase, resulting in a wider and more spread out central maximum. This is because the amount of diffraction depends on the ratio of wavelength to the distance between the diffracting elements, and the wavelength of yellow light is greater than blue light.

26. The optical phenomena, twinkling of stars is due to

(a) atmospheric reflection

(b) total reflection

(c) atmospheric refraction
(d) total refraction
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

The phenomenon responsible for twinkling of stars is refraction. Light from the stars has to come through a thick layer of atmosphere and also the density of atmosphere keeps on changing as gravitation increases so more dense layer of atmosphere will have greater refractive index and hence will bend light more. Since the physical condition of atmosphere is not constant and keeps on changing so the amount of star light entering our eye keeps on changing thus sometimes they appear bright while sometimes faint.

27. We can not see in the fog, because

(a) fog absorbs light
(b) the refractive index of fog is infinite
(c) there is complete reflection of light on fog drops
(d) light becomes scattered by fog drops
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

Atoms and molecules in the air, including anything carried in the air like dust or smoke, will scatter light. Water droplets, as they are present in fog, also scatter light. The light falling on an object and reflected to a viewer can be scattered to heck and back before it gets to the place where it can be 'seen' by an observer. So the observer just sees a 'whiteout' instead of being able to make out anything beyond a few meters or so.

28. Twinkling of stars during the night time can be explained with -

(a) Refraction of light (b) Reflection of light
(c) Polarization of light (d) Interference of light
(c) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

The reason for the twinkling of stars during the night time is the refraction of the light of the stars.

29. For shaving, one uses –

(a) Concave mirror (b) Plain mirror
(c) Convex mirror (d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (a)

People use a concave mirror for shaving because when a man standing between the principal focus and pole of a concave mirror, he notices an enlarged, erect and virtual image of his face. This is the reason why a concave mirror of a large focal length is used for shaving.

30. Light is made of seven colours. What is the method of separating the colours?

(a) The colours can be separated by a prism
(b) The colours can be separated by a filter
(c) The colours can be separated by plants
(d) The colours cannot be separated

47th B.P.S.C. (Pre) 2005

Ans. (a)

The colours can be separated by using a prism. The white colour is made up of seven colours. When it passes through a prism, due to different wavelengths, different colours of light are refracted by a different amount.

31. Dispersion of light is possible by :

(a) Prism (b) Convex lens
(c) Concave lens (d) Simple mirror
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Dispersion of light is possible by prism. On passing through the prism, the white light is separated into its different component colours, which is known as dispersion.

32. Which colour has the maximum deviation in the dispersion of white light passing through the prism?

(a) Green (b) Red
(c) Violet (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

In the dispersion of white light passing through the prism, violet has the maximum deviation as it has least wavelength among all the colors, it bends most upon incidence and has maximum deviation

33. The coloured light that refracts most while passing through a prism is

(a) yellow (b) violet
(c) blue (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Violet light is refracted the most while passing through a prism because violet colour has the minimum speed in prism and violet have shortest wavelength among VIBGYOR. Because of its shortest wavelength, it bends most upon incidence and has maximum deviation.

34. Sea seems blue due to –

(a) Excess deepness

(b) Reflection of sky and scattering of light by the drops of water

(c) Blue colour of water

(d) Upper layer of water

40th B.P.S.C. (Pre) 1995

Ans. (b)

The blue appearance of the sky is due to scattering of sunlight from atmosphere. When we look at the sky, it is the scattered light which enters the eyes. Among the shorter wavelengths, the blue colour is present in a larger proportion in sunlight. Light of shorter wavelength is scattered by air molecules which because of their smaller size follow rayleigh's scattering. Blue light is strongly scattered by the air molecules and reach the observer. This explains the blue colour of the sky. This phenomena seen like this sea are blue colour.

35. Though water is transparent to visible light, it is not possible to see distant objects in fog which consists of fine drops of water. This is so because

(a) fine drops of water are opaque to visible light

(b) most of the light is scattered to create apparent opacity

(c) light rays suffer total internal reflection and so unable to reach observer's eyes

(d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (b)

Though water is transparent to visible light, it is not possible to see distant objects in fog which consists of fine drops of water. This is so because most of the light is scattered to create apparent opacity.

36. In which direction the rainbow is seen at 12 noon?

(a) In the West (b) In the South

(c) In the East (d) It cannot be seen

43rd B.P.S.C. (Pre) 1999

Ans. (d)

A rainbow is always formed in the opposite direction to the sun, so that sunlight striking water droplets (which act as prism) can be easily refracted and dispersed, resulting in a rainbow. It occurs when light is reflected, refracted and dispersed in water droplets, resulting in a spectrum of light seen in the sky. A rainbow is located opposite to sun; and at 12 noon. Sun is at overhead position that's why rainbow are not seen at noon.

37. To remove the defect of long sightedness one uses –

(a) Concave lens

(b) Convex mirror

(c) Convex lens

(d) Concave mirror

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Far-sightedness (hypermetropia) as it is medically termed is a vision condition in which distant object are usually seen clear, but near objects are not clearly visible. To remove this vision problem one should use a convex lens.

38. Reading glasses are made from which type of lenses?

(a) Concave (b) Convex

(c) Plain (d) None of these

44th B.P.S.C. (Pre) 2000

Ans. (b)

A convex lens is thicker in the middle and thinner at the edges. Rays of light that pass through the lens are brought closer together. A convex lens is also called a converging lens. A convex lens is used in reading glasses and it also used to remove the defect of far-sightedness.

39. Large number of thin strips of black paint are made on the surface of a convex lens of focal length 20 cm to catch the image of a white horse. The image will be

(a) a horse of less brightness

(b) a zebra of black stripes

(c) a horse of black stripes

(d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

When capturing an image with a camera, light rays from the object pass through the lens to form the picture, if the lens is partially covered, the image can still be formed, but the overall brightness of the resulting photograph will be reduced compared to an image taken without any obstruction on the lens.
That's why the image of the white horse will be of less brightness.

40. The power of lens is measured in :

(a) Watt (b) Ampere

(c) Volt (d) Diopter

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)